

Rocket Elevators

ELEVATE SAFETY, SERVICE AND VALUE

INTERNAL DOCUMENT

Common Failure Modes by Elevator Type

Internal document. Last updated April 2026. Contact: Field Operations Manager, Rocket Elevators Ontario Division.

How to Use This Document

This document maps common failure modes across our fleet of approximately 40,000 elevating devices in Ontario. It is organized by system first (door, leveling, hydraulic, traction, electrical, safety devices), then by device type. For each failure mode: what it looks like, what causes it, risk level, and recommended action.

Risk levels:

- **Low:** cosmetic or inconvenience; no safety concern
- **Moderate:** operational impact or early indicator of a developing problem
- **High:** direct safety risk or regulatory compliance concern
- **Critical:** immediate danger to life; requires emergency response and device shutdown

Our three contractors handle different segments of the fleet. **ThyssenKrupp Elevator** services traction and MRL installations. **KONE** handles hydraulic systems and modernizations. **Schindler** manages specialty and observation elevators. Field technicians handle minor repairs directly, but anything affecting safety circuits, structural integrity, or rated capacity must be escalated to the appropriate contractor.

Door System Failures

Door-related service calls account for roughly 40% of all calls across our fleet. Doors are the most mechanically active component on any elevator, cycling thousands of times per day on busy installations, and they produce the most incidents (strikes, entrapments, near-misses) of any system.

Door Operator Failures

The **door operator** is the motor and drive assembly that opens and closes the car doors. Failure modes:

- **Motor burnout:** overheats and fails after years of cycling. Signs include sluggish movement, humming, or burning smell.
- **Belt or chain wear:** stretches, slips, or breaks. Doors hesitate, move unevenly, or fail to open fully.
- **Clutch issues:** the clutch coupling the car door to the landing door wears, causing partial openings.
- **Drive board failure:** electronic drive boards control speed and force. Board failures cause erratic behavior (slamming, creeping, oscillating).

Risk level: moderate. Failed operators cause entrapment or service disruption. Erratic behavior can cause door strikes.

Action: field technicians can replace belts, adjust clutches, and reset drive boards. Motor and board replacement require a contractor.

Door Interlock Failures

Door interlocks prevent the car from moving unless all landing doors are fully closed and locked. An interlock failure can allow the car to move with a door open, exposing passengers to the hoistway.

- **Mechanical wear:** the locking mechanism wears from repeated cycling, eventually failing to engage fully.
- **Electrical contact degradation:** contacts corrode, pit, or accumulate debris, causing intermittent faults.
- **Misalignment:** building settlement or door track shifting moves the interlock out of alignment with its striker.

Risk level: high to critical. Interlock failures are a leading cause of serious elevator accidents. Ontario Regulation 209/01 requires annual interlock testing.

Action: always escalate to a contractor. If an interlock failure is suspected, take the device out of service immediately.

Door Tracking and Alignment

Door panels ride on **rollers** along a **track** (header). Over time:

- Rollers wear flat, causing the door to drag, wobble, or bind
- Tracks deform from impact (loading equipment hitting doors on freight elevators) or from accumulated debris
- **Hangers** (the brackets connecting door panels to rollers) crack or bend, causing the panel to hang at an angle

Misaligned doors create gaps, bind against the frame, or fail to close fully. On freight elevators, tracking damage is accelerated by forklifts and heavy carts striking the doors during loading.

Risk level: moderate. Tracking issues usually result in doors that won't close (causing the car to stop responding) rather than an unsafe condition. However, a door panel that detaches from its hanger is a serious hazard.

Closing Force Issues

CSA B44 limits the kinetic energy a closing door can deliver. Doors that close too hard cause injuries (our incident data includes numerous door strikes on arms, hands, fingers, and feet). Doors that close too softly may not fully close, preventing the interlock from engaging.

Closing force drifts over time as components wear and should be checked during routine maintenance.

Risk level: moderate to high. Door strikes are the single most common incident type in our fleet.

Reopening Device Failures

Reopening devices (infrared curtains, light beams, or mechanical safety edges) detect obstructions and reverse closing doors. Failures:

- **Infrared sensor failures:** sensors go blind from dirt, sun glare, or component failure. The door closes without detecting a person.
- **Mechanical safety edge failures:** the rubber edge hardens with age, loses sensitivity, or disconnects from its switch.
- **Nuisance reversals:** a malfunctioning sensor detects phantom obstructions, causing doors to cycle without closing.

Risk level: high when the device fails to detect obstructions. Failed reopening devices are a common TSSA inspection citation.

Action: field technicians can clean sensors and adjust mechanical edges. Sensor replacement requires a contractor.

Sill Wear and Gap Issues

The **sill** is the threshold at the bottom of the door opening. Sills wear down from foot traffic and rolling loads, creating gaps where objects (shoes, cane tips, small wheels, fingers) can get caught. CSA B44 specifies maximum allowable gaps.

Risk level: moderate. Sill gaps develop gradually and are caught during routine inspections.

Freight Elevator Door and Gate Failures

Freight elevators use door configurations not found on passenger units:

- **Vertical-biparting doors:** two panels meeting in the middle. The counterbalance system for the upper panel fatigues over time, causing the panel to drop too fast or not stay open.
- **Manual gates with interlocks:** gate latches wear, interlocks corrode, and operators learn to force the gate, potentially bypassing the interlock.
- **Power-operated gates:** same operator failures as passenger doors, with added stress from gate weight and rougher operating environment.

Risk level: high for interlock bypass on manual gates. Moderate for other freight door failures. See the Safety Code Quick Reference for CSA B44 requirements on door and gate interlocks.

Leveling and Positioning Failures

Leveling failures are the second most common cause of incidents in our fleet. **Out of level** means the car floor does not align with the landing floor when the doors open. A properly functioning elevator should level to within one-quarter inch. Anything beyond half an inch is a trip hazard; beyond two inches is a fall hazard. Our incident data shows differences ranging from 1 inch to over 12 inches, with even 2 to 3 inch gaps causing broken wrists, ankle injuries, knee injuries, and head injuries.

Leveling in Traction Elevators

Traction elevators position the car using an **encoder** (or, in older systems, a selector) that tracks car position. Leveling failures stem from:

- **Encoder drift:** loses calibration over time, especially after power outages. The controller thinks the car is level when it is off by inches.
- **Brake timing issues:** brake applies too early (stops short) or too late (overshoots). Wear changes timing gradually.
- **Rope stretch:** ropes stretch under load. A car that levels perfectly when empty may be 2 to 3 inches low fully loaded. Newer controllers compensate; older ones do not.
- **Load-dependent errors:** compensation chain/rope issues on taller installations cause leveling to vary by floor.

Leveling in Hydraulic Elevators

Hydraulic elevators position the car by controlling fluid flow. Leveling failures include:

- **Valve drift:** control valve does not seat perfectly, allowing fluid to leak past. The car settles slowly after stopping, sometimes dropping an inch or more over minutes.
- **Seal wear:** cylinder seals degrade, allowing fluid to bypass the piston. Same settling behavior as valve drift, but the root cause is in the cylinder.
- **Temperature effects:** fluid viscosity changes with temperature. Levels perfectly in summer but overshoots in winter (thicker fluid) or undershoots in extreme heat (thinner fluid).
- **Fluid level changes:** low fluid levels alter system response. The car may level differently at different floors.

Age and Leveling

Leveling accuracy degrades with age across all elevator types. Older controllers use less precise positioning, and worn components introduce variability.

Risk level: high. Leveling failures cause real injuries.

Frequency: complaints increase significantly in devices over 20 years old. Chronic leveling issues in devices over 30 are a strong indicator that modernization should be discussed with the building owner.

Action: field technicians can report leveling measurements and perform basic adjustments. Persistent leveling issues require contractor evaluation and may indicate the need for controller replacement.

Hydraulic System Failures

Hydraulic elevators make up a significant portion of our low-rise fleet: passenger elevators under six stories, Freight Elevator-P units, some LULA elevators, and sidewalk elevators. Hydraulic failures are typically gradual rather than catastrophic, often detected during routine maintenance before becoming dangerous. However, they can be expensive and create environmental concerns (oil contamination).

Seal and Packing Failures

The **packing** (seals around the hydraulic jack) prevents oil from leaking past the piston. Over time, packing hardens, cracks, or wears. Signs: oil stains on the cylinder or in the pit, the car slowly drifting down when

stationary, and fluid levels requiring periodic top-ups. Packing replacement typically requires one to two days out of service.

Hydraulic Fluid Degradation

Hydraulic fluid degrades through **contamination** (dirt, metal particles, moisture accelerating component wear), **viscosity breakdown** (affecting leveling accuracy and valve response), and **water intrusion** (from condensation, leaking fittings, or pit flooding, causing corrosion and cavitation). Fluid analysis is part of routine maintenance.

Pump and Motor Failures

The **hydraulic power unit** (pump and motor assembly) pressurizes the fluid. Failure modes:

- **Overheating:** continuous operation without adequate cooling. May trigger thermal protection and shut down the elevator.
- **Cavitation:** air in hydraulic lines causes loud knocking or whining, damages pump internals. Often caused by low fluid levels or suction line air leaks.
- **Bearing wear:** causes vibration, noise, and eventually seizure.

Control Valve Failures

The **control valve** regulates fluid flow to control car speed and stopping position. Valve failures include:

- **Stuck valve:** the valve spool or poppet seizes in position. The car either won't move (stuck closed) or won't stop properly (stuck open, though safety valves prevent free-fall).
- **Leaking valve:** the valve does not seat fully, allowing fluid to bypass. The car drifts down slowly when it should be stationary. This is the most common valve failure and a frequent source of leveling complaints.
- **Solenoid failure:** the electrical solenoid that actuates the valve burns out or loses its magnetic strength. The valve fails to respond to controller commands.

Underground Cylinder Corrosion

Many older hydraulic elevators use **in-ground cylinders** that extend into a bore hole beneath the pit. In direct contact with soil and groundwater, they corrode over time, leading to fluid loss into the ground (an environmental contamination issue), cylinder wall thinning, and inconsistent car movement. Ontario environmental regulations require reporting fluid releases into soil. Remediation options (cylinder lining, replacement, or conversion to holeless) are all major alterations requiring TSSA approval. This is a fleet-wide concern for our older hydraulic installations.

Hose Failure

High-pressure hydraulic hoses connect the power unit to the cylinder. Burst hose failure is rare but dangerous (hot, high-pressure oil release). Hoses should be inspected for cracking, bulging, and fitting corrosion, and replaced on a schedule.

Risk level: moderate overall. The exceptions are hose burst and in-ground cylinder failure.

Action: field technicians can check fluid levels, inspect for leaks, and report abnormal sounds. All hydraulic repairs beyond basic monitoring require a contractor, typically KONE for our fleet. See the Hydraulic Elevator Maintenance Procedures for preventive maintenance schedules.

Traction System Failures

Traction elevators serve our mid- and high-rise installations: passenger elevators, Freight Elevator-E units, and observation elevators. ThyssenKrupp Elevator handles most traction maintenance, including MRL installations.

Rope Wear

Hoist ropes (wire ropes) suspend the car and counterweight, typically 4 to 8 ropes in parallel. Failure modes:

- **Broken wires:** individual wires break from fatigue. CSA B44 specifies the maximum broken wires per lay length before replacement.
- **Diameter reduction:** rope thins as wires break and core compresses.
- **Corrosion:** moisture from pit flooding, roof leaks, or condensation weakens rope wires.
- **Unequal tension:** unevenly stretched ropes overload some strands, accelerating wear and causing vibration.

Risk level: high. Rope failure is extremely rare on a properly maintained elevator (multiple ropes provide redundancy), but the consequences are catastrophic. Rope condition is one of the most monitored items in our inspection program.

Sheave Groove Wear

The **sheave** (drive pulley) has grooves that the ropes ride in. Over time, the grooves wear and change shape:

- **Undercut grooves:** the bottom of the groove wears deeper than intended, reducing the contact angle and the friction that drives the car. Ropes can slip on an undercut sheave, especially under heavy load.
- **Uneven groove wear:** grooves wear at different rates, causing ropes to ride at different heights. This creates unequal rope tension and vibration.

Sheave regrooving or replacement is a major job, typically requiring the car to be out of service for several days. It is a contractor operation.

Motor and Drive Failures

Modern traction elevators use **variable-frequency drives (VFDs)** to control the motor. Older installations may use motor-generator sets or DC drives. Failure modes:

- **Overheating:** inadequate ventilation, dust on heat sinks, or failed cooling fans. Thermal protection shuts the elevator down.
- **Bearing failure:** causes vibration and noise; if unaddressed, can damage the motor shaft or windings.
- **Drive electronics:** VFD boards fail from age, heat, or power quality issues. Usually the car stops safely, but is out of service until repaired.
- **Encoder problems:** encoder on the motor shaft provides speed/position feedback. Failure causes erratic movement, leveling errors, or shutdown.

Brake Failures

The **machine brake** holds the car in position when the motor is not running. It is fail-safe: springs apply the brake, an electromagnet releases it. Failure modes:

- **Lining wear:** pads wear down, reducing braking force. Car may drift.
- **Spring fatigue:** springs lose tension, meaning insufficient braking force.
- **Solenoid problems:** electromagnet fails, either holding the brake on permanently (car won't move) or failing to hold it (brake drags or allows unintended movement).

Risk level: critical. Brake failure can result in uncontrolled car movement. Tested every maintenance visit, inspected annually by TSSA.

Action: brake adjustment is a contractor operation. Field technicians should report any signs of brake problems immediately.

Counterweight Issues

The **counterweight** balances the car weight plus approximately 40% of rated load. Guide shoe wear causes wobble and noise. Buffer condition (oil levels for oil buffers, spring integrity for spring buffers) must be maintained.

Gearbox Failures (Geared Traction Only)

Older traction elevators use a **gearbox** between motor and sheave (less common in new installations, but our fleet includes many). Failure modes: oil leaks from degraded seals, gear tooth wear causing noise and vibration, and bearing failure.

Risk level: varies. Rope and brake issues are high to critical. Motor, drive, and gearbox issues are usually moderate (car stops safely rather than moving unsafely). See the Traction Elevator Troubleshooting Guide for symptom-based diagnosis.

Electrical and Controller Failures

The **controller** processes calls, manages car movement, monitors safety circuits, and coordinates with other elevators in a group. Electrical and controller failures increase significantly with age and are a primary driver of modernization decisions.

Relay Controller Failures

Older elevators (pre-1990s) use **relay-based controllers** with hundreds of electromechanical relays. Failure modes: **contact welding** (relay sticks, potentially bypassing safety logic), **coil burnout** (relay stops responding), and **timing drift** (affects door timing, leveling, group dispatch). Relay controllers are increasingly difficult to maintain because replacement parts are becoming scarce. When failures are frequent and parts hard to find, modernization is the practical path forward.

Microprocessor Controller Failures

Modern **microprocessor-based controllers** are more reliable but have their own failure modes:

- **Board failures:** circuit boards fail from aging, heat, moisture, or power events.

- **Software faults:** controller enters unexpected states after power glitches. Resets often resolve the issue; recurring faults indicate a deeper problem.
- **Memory corruption:** non-volatile memory storing configuration parameters corrupts over time, causing erratic behavior.

Power Quality

Ontario's power grid is generally reliable, but **voltage sags** (cause controller resets, drive faults), **phase loss** (older controllers may attempt to run on reduced power, damaging the motor), and **surges** (lightning, switching events) all affect elevator operation. Surge protection on the elevator's power feed is essential but not always present in older installations.

Wiring Degradation

Wiring degrades over time: **insulation breakdown** (becomes brittle, cracks, exposing conductors), **loose connections** (from vibration and thermal cycling, creating hard-to-diagnose intermittent faults), and **rodent damage** (a recurring problem in buildings with food service, loading docks, or older construction).

Traveling Cable Wear

The **traveling cable** connects the car to the controller, flexing with every trip. Over time, conductors break at flex points, the jacket cracks, and individual circuits become intermittent (communication dropouts, indicator failures, intercom problems). Replacement requires a full day out of service.

Risk level: usually moderate. Most electrical failures result in the car stopping safely. The exception is safety circuit failures, where a wiring fault could bypass a safety device.

Frequency: controllers under 15 years rarely have issues. Between 15 and 25, failure rates increase. Over 25, controllers are typically past their useful life and should be prioritized for modernization.

Safety Device Failures

Safety devices are the last line of defense, activating only in emergencies (overspeed, free-fall, buffer impact). They must work perfectly when called upon, even though they may go years without activating. This means failures are typically discovered during testing, not during incidents.

Governor

The **governor** monitors car speed via a rope running the length of the hoistway. When the car exceeds a preset overspeed threshold, the governor trips, activating the car safety device. Failure modes:

- **Rope wear:** wears from friction, fatigue, and environmental exposure
- **Switch calibration drift:** triggers at too high a speed (failing to protect) or too low (nuisance trips)
- **Lubrication:** dry pivots and linkages can seize, preventing the governor from tripping

Governors must be tested annually per CSA B44.

Car Safety Device (Safeties)

The **car safety device** ("the safeties") grips the guide rails to stop the car in a free-fall or overspeed condition, activated by the governor. Failure modes:

- **Seizure from disuse:** mechanism seizes from corrosion, dried lubricant, or debris
- **Wear on gripping surfaces:** jaws or rollers wear from vibration and environmental exposure
- **Linkage failures:** the connection from governor rope to safety device can corrode, stretch, or disconnect

Annual testing is mandatory. After a full-speed application (rare, typically only in a real emergency), the device must be inspected and recertified.

Buffers

Buffers absorb the car's energy if it overruns the lowest landing. **Spring buffers** (low-speed elevators) fatigue over time. **Oil buffers** (higher-speed) require correct fluid level and viscosity. Buffer condition is checked during routine maintenance and annual inspections.

Overspeed Protection

Modern elevators add **unintended car movement (UCM) protection** (detects the car moving without a valid run command) and **ascending car overspeed protection (ACOP)** (prevents up-direction overspeed). Both are required on newer installations per CSA B44 amendments.

Risk level: critical. Safety device failure during an emergency can result in serious injury or death. The annual test program is essential; missing or deferring safety device tests is never acceptable. See the Safety Code Quick Reference for CSA B44 testing requirements.

Type-Specific Failure Patterns

Each device type has characteristics that shift failure frequency and severity. This section covers the patterns we see most often by type.

Passenger Elevator

Approximately 72% of our fleet. Highest use cycles of any device type.

- **Door components wear fastest** due to sheer cycle count. A busy commercial elevator may cycle its doors 500+ times per day.
- **Cab finish damage** from heavy traffic (floors, walls, ceiling). While cosmetic, deteriorated finishes affect building owner relationships.
- **Button and indicator failures** from volume of use and vandalism.
- **Emergency phone failures** from component aging, traveling cable degradation, and vandalism. Monthly phone testing is a regulatory requirement.

Passenger elevators exist in both traction and hydraulic configurations.

Freight Elevator

Freight elevators (Freight Elevator-E for electric traction, Freight Elevator-P for hydraulic) account for approximately 15% of our fleet.

- **Overloading is the primary accelerator of wear.** Users frequently exceed rated capacity (a 5,000 lb elevator regularly seeing 6,000-7,000 lbs). Overloading stresses every component and is the single biggest contributor to premature failure.
- **Door and gate damage from loading equipment.** Forklifts, pallet jacks, and carts collide with doors routinely. Door systems may need repair at two to three times the passenger rate.
- **Pit contamination.** Debris, spilled materials, and water damage pit equipment.
- **Stringent leveling requirements.** Even a one-inch difference is a serious problem for forklifts, which can tip or lose their load.
- **Sill damage.** Heavy rolling loads wear sills far faster than foot traffic.

Observation Elevator

Smaller fleet segment, managed primarily by Schindler. Defining characteristic: environmental exposure.

- **Weather exposure.** Exterior-mounted units face rain, snow, ice, wind, and temperature extremes. Water intrusion is an ongoing challenge.
- **Glass panel issues.** Seal failures (water intrusion), cracking from thermal stress, delamination. Replacement requires specialized skills.
- **Exterior guide rail corrosion.** Exposed rails corrode faster, affecting ride quality and safety device engagement.
- **Increased maintenance frequency** and higher per-unit costs due to environmental exposure.

LULA Elevator

Limited Use/Limited Application (LULA) elevators are small, low-speed units for accessibility where a full-size elevator is not feasible.

- **Motor and drive burnout from unanticipated use.** Designed for limited trips per hour; if building use changes (residential to clinic), the LULA is used beyond its design, causing overheating and drive failures.
- **Simpler safety systems** with less redundancy. Less margin for component degradation before a safety impact.
- **Accessibility component wear.** Smaller, less robust door operators. Low speed can mask leveling issues.

Sidewalk Elevator

Sidewalk elevators move goods between street level and a basement, with a hatch at grade level.

- **Hatch seal failures.** Must keep out rainwater, snowmelt, and debris. Seal failure floods the hoistway and pit.
- **Public exposure.** Subject to vandalism, interference, vehicles driving over hatches, and street debris.
- **Unique safety concern.** If the hatch opens unexpectedly, pedestrians can fall into the opening. Hatch interlocks and guarding are critical.

Power Type Manlift

Legacy continuous-loop, open-platform devices. We are actively phasing these out.

- **Step and belt wear.** Worn steps create trip hazards; worn belts can slip or break.
- **Guarding failures.** Guards at each landing prevent riders from being carried into the overhead structure. Damaged or missing guards create a crushing hazard.
- **Inherent design risk.** No safety device comparable to a governor and safeties. If a rider falls, the device does not stop.

Our standard recommendation is replacement with a passenger elevator or LULA. TSSA has become increasingly stringent about manlift compliance, and insurers are raising rates or declining coverage.

Material Lift - ATD and Special Installations

Material Lift - ATD (Automatic Transfer Device) units are industrial lifts in warehouse environments. **Special Installations** are non-standard configurations. Both are low-count fleet segments. Material lifts face overloading similar to freight elevators but with less robust construction. Special installations require individual assessment; standard procedures may not apply.

Temporary Elevator

Construction site elevators, removed when permanent elevators are commissioned. Exposure to dust, debris, concrete slurry, and weather. Rough use, temporary wiring. Short service life means acute construction damage is the primary concern.

Age-Related Patterns and Modernization Indicators

Age is the single strongest predictor of failure frequency across all device types. The general pattern:

Under 15 years: mostly routine maintenance. Component failures are isolated. Door components and cab finishes are the most common service items. Service call frequency is low and predictable.

15 to 25 years: controllers and electrical systems begin to age out. Door components have been through replacement cycles. Controller boards become harder to source. Service call frequency increases, shifting from routine maintenance to component replacement. This is the window where proactive planning pays the biggest dividends.

25 years and older: multiple systems at or past design life. Controllers may be on obsolete platforms. Parts availability becomes a real constraint. Compliance with current CSA B44 may require alterations. Energy consumption increases. Service call frequency is high, and emergency calls (entrapments, out-of-service) become more common. At this age, cumulative repair costs often exceed modernization cost.

Signals That Suggest Modernization

Operations managers should initiate a modernization discussion with the building owner when a device shows multiple indicators from this list:

- Service call frequency has increased by 50% or more over the past two years
- Multiple compliance orders on the same inspection, especially for the same components
- Parts are on extended lead times (weeks or months) or are no longer manufactured

- The controller platform has been discontinued by the manufacturer
- Safety device testing reveals degraded performance that is correctable but recurring
- Energy consumption is significantly higher than comparable modern equipment
- Ride quality complaints are increasing (noise, vibration, rough stops)
- The device has accumulated multiple major alterations, each addressing a different system, rather than a coordinated modernization

Our fleet includes a substantial number of devices in the 25-plus year range. A well-timed modernization conversation, supported by maintenance history and cost data, is far more productive than waiting for a catastrophic failure or TSSA shutdown order.

Failure Mode Impact on Risk Predictions

The failure modes described in this document connect directly to the risk prediction model's scoring factors.

Failed inspections are a primary input. Recurring failures for safety-critical components (interlocks, safety devices, brakes, ropes) receive higher scores. The model distinguishes isolated failures from patterns: three failed inspections in five years for the same component signals a systemic issue.

Compliance orders carry weight based on nature and recurrence. A cosmetic compliance order (cab finish, indicator lamp) has minimal impact. A safety device or interlock compliance order has significant impact. Recurring orders for the same component signal an unresolved problem.

Past incidents inform risk profiles. Devices with entrapment, door strike, or leveling injury histories score higher, even when incidents were addressed. The rationale: incident history reflects both mechanical condition and operating environment, and environmental factors persist after repairs.

Age is factored in combination with maintenance history, inspection results, and incident patterns. A well-maintained 30-year-old elevator with clean inspections may score lower than a poorly maintained 15-year-old with multiple compliance orders.

A device showing multiple failure modes (recurring leveling issues plus door problems plus aging controller plus compliance orders) is very likely to appear in the high-risk tier. When a device is flagged, refer to this document to understand which failure modes are likely contributing.

See Emergency Response Protocols for responding to equipment failures that create immediate safety concerns. See the operations handbook for details on how compliance orders are scored in the risk model.